

Chapter 5

Image Compression

IMP302m

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Fundamentals

5.1 Lossless coding

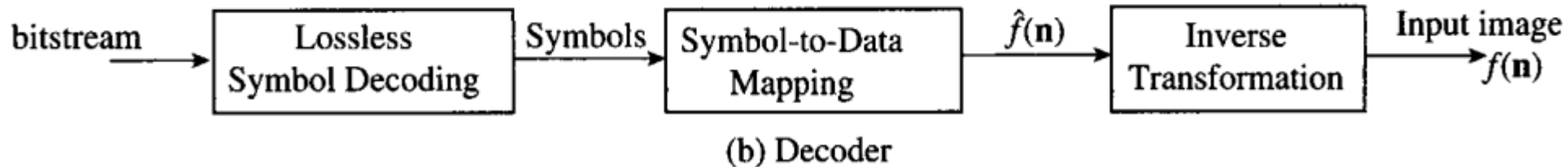
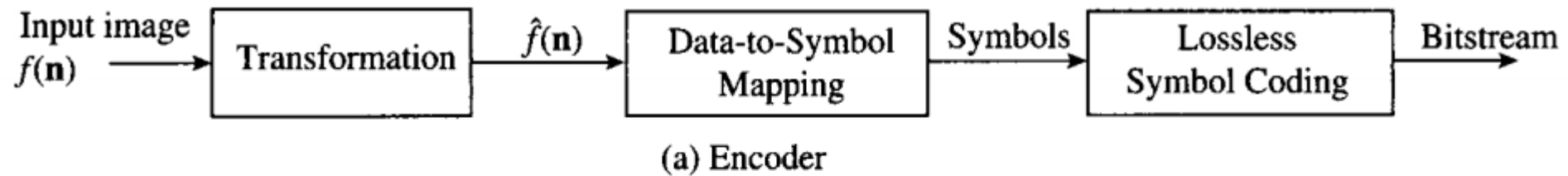
- The goal of lossless image compression is to reduce the number of bits needed to represent an image without losing any information, thereby improving transmission speed and minimizing storage.
- measured as an average bit-rate
- lossy compression aims to balance image quality and bit-rate, allowing some information loss to achieve a greater reduction in bit-rate
 - Need for real-time applications
- The process of compression is commonly referred to as coding

5.1 Lossless coding

- Lossless compression is possible due to the redundancy in image signals, which is linked to the correlation among image data samples.
 - In natural still images: high spatial correlation between neighboring samples
 - In video: additional temporal correlation between samples in successive frames.
- In lossless coding, the decoded image must be identical to the original in both numerical and visual aspects, but this limits compression to a factor of two or three
- To achieve higher compression, perceptually lossless coding removes both redundant and perceptually irrelevant information, allowing some loss as long as the recovered image appears visually identical to the original, even if it's not numerically the same.

5.1 Lossless coding – Basics of lossless image coding system

- The block diagram of a lossless coding system



5.1 Lossless coding – Basics of lossless image coding system

- Transformation: convert the input image data $f(n)$ into a form $\hat{f}(n)$ that can be compressed more efficiently.
 - differential/predictive mapping, unitary transforms such as DCT, subband decompositions
- Data to Symbol Mapping: convert the image data into entities called symbols that can be efficiently coded by the final stage
 - Partitioned into blocks or not
 - Important for optimizing the coding
- Lossless Symbol Coding: generate a binary bitstream by assigning binary codewords to the input symbols

5.1 Lossless coding – Coding selection

- Selecting which lossless coding method to use depends on
 - the application and usually involves a tradeoff between several factors including the implementation hardware or software,
 - the allowable coding delay,
 - and the required compression level.
- Factors
 - Compression efficiency: compression ratio compares the size of the original input image data with the size of the generated compressed bitstream
 - Code delay: minimum time for both encode and decode an input data sample
 - Implement complexity: the number of arithmetic operations and memory requirements
 - Robustness: how coding method is affected by error environment
 - Scalability: generates layered bitstream embedding a hierarchic representation of the input image data

5.1 Lossless coding – Basic concepts from information theory

- A source include N symbols $S = \{s_0, s_1, \dots, s_{N-1}\}$
- The probability that the source outputs the symbol s_k at any time is p_k

$$\sum_{k=0}^{N-1} p_k = 1$$

- Information associated with a symbol (self-information)

$$I_{s_k} = -\log_2(p_k) \text{ (bits)}$$

- Source entropy (first-order entropy or marginal entropy)

$$H(S) = \mathbb{E} \{I_k\} = - \sum_{k=0}^{N-1} p_k \log_2(p_k) \text{ (bits per symbol)}$$

5.1 Lossless coding – Basic concepts from information theory

- Average bit-rate corresponding to code C

$$B_C = \sum_{k=0}^{N-1} p_k l_k \text{ (bits per symbol)}$$

where l_k is the number of bits of the symbol s_k

- For any uniquely decodable lossless code C , the entropy $H(S)$ is a lower bound on the average bit-rate B_C

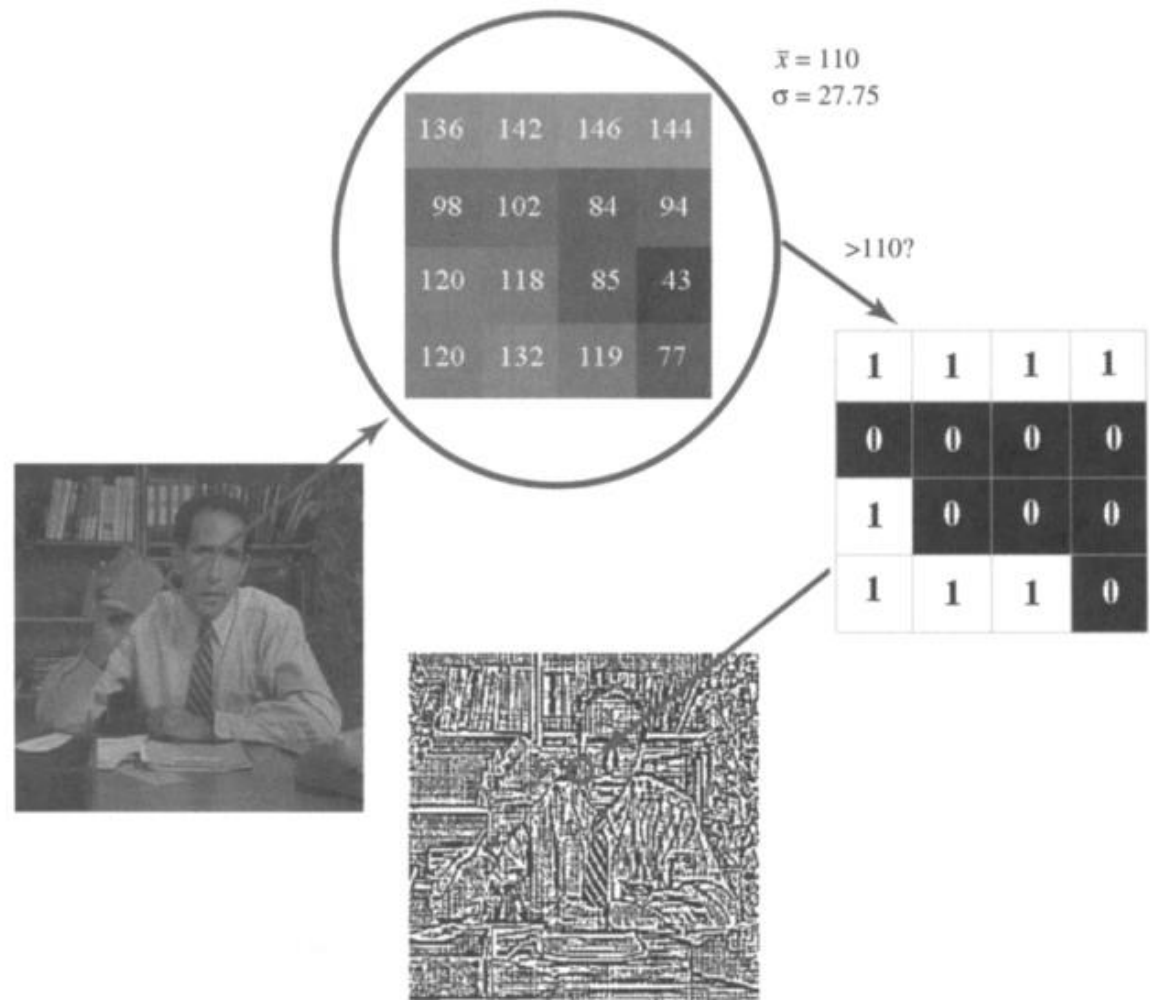
$$B_C \geq H(S)$$

5.2 Block Truncation Coding (BTC)

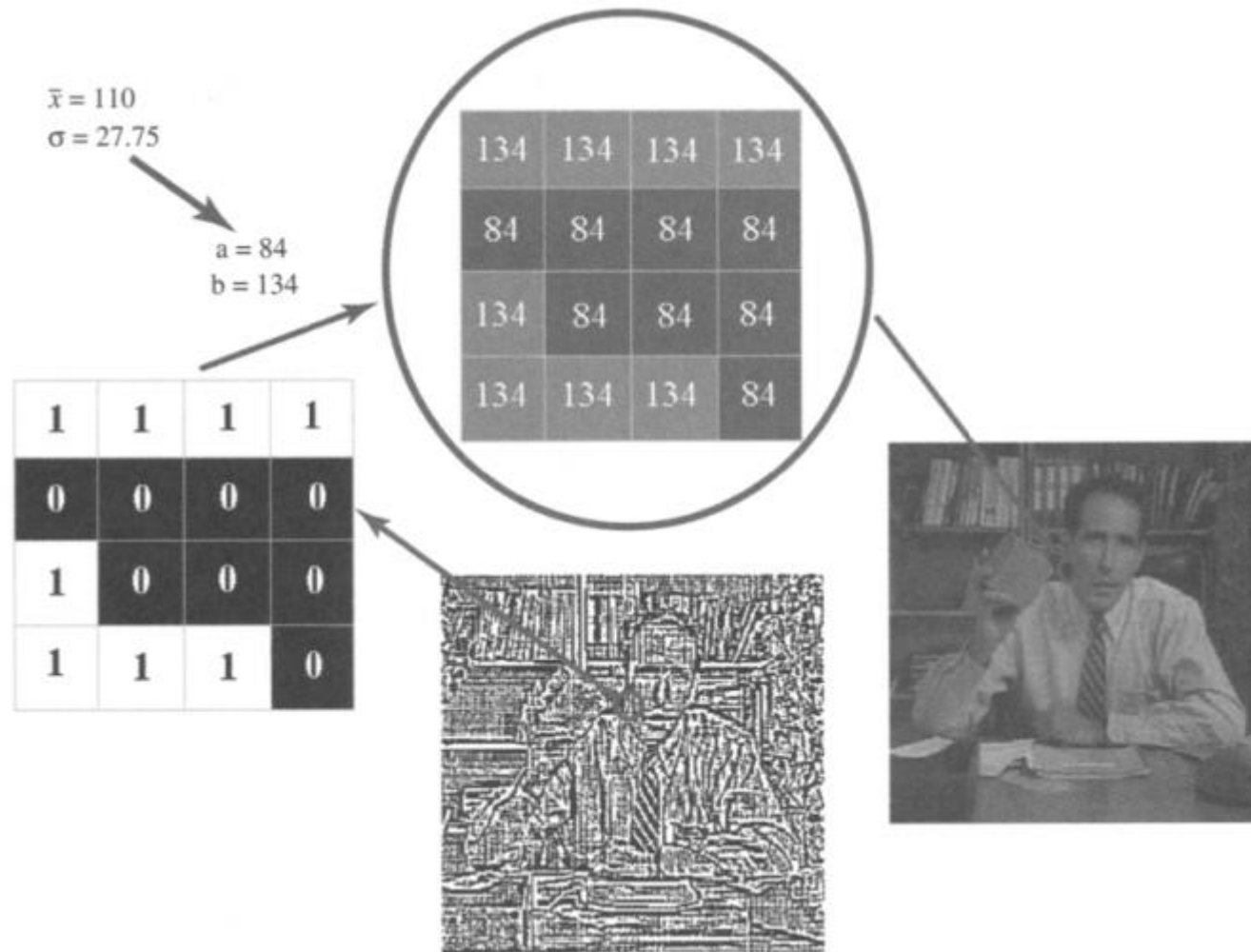
- Lossy fixed length compression technique that uses a Q level quantizer to quantize a local region of the image.
- A number of the moments of a local region in the image are preserved in the quantized output.
 - Simplest form: preserve the sample mean and sample standard deviation of a grayscale image.

5.2 Block Truncation Coding (BTC) – Encode (compression)

- For a two level (1 bit) quantizer, select two luminance values a and b which are used for decoding



5.2 Block Truncation Coding (BTC) – Decode (decompression)



5.2 Block Truncation Coding (BTC) – How to obtain a and b

- Consider k pixels of $n \times n$ block ($k = n \times n$) and the intensity values of the pixels are $x_1, x_2, \dots, x_k$
- Two sample moments

$$m_1 = \bar{x} = \frac{1}{k} \sum_{i=1}^k x_i$$

$$m_2 = \frac{1}{k} \sum_{i=1}^k x_i^2$$

- The sample standard deviation $\sigma^2 = m_2 - m_1^2$
- q the number of pixels in block $\geq x_{th}$

$$\text{Output} = \begin{cases} b & \text{if } x_i \geq x_{th} \\ a & \text{if } x_i < x_{th} \end{cases}$$

$$\begin{cases} a = m_1 - \sigma \sqrt{\frac{q}{k-q}} \\ b = m_1 + \sigma \sqrt{\frac{q}{k-q}} \end{cases}$$

5.2 Block Truncation Coding (BTC) – Variations and Applications

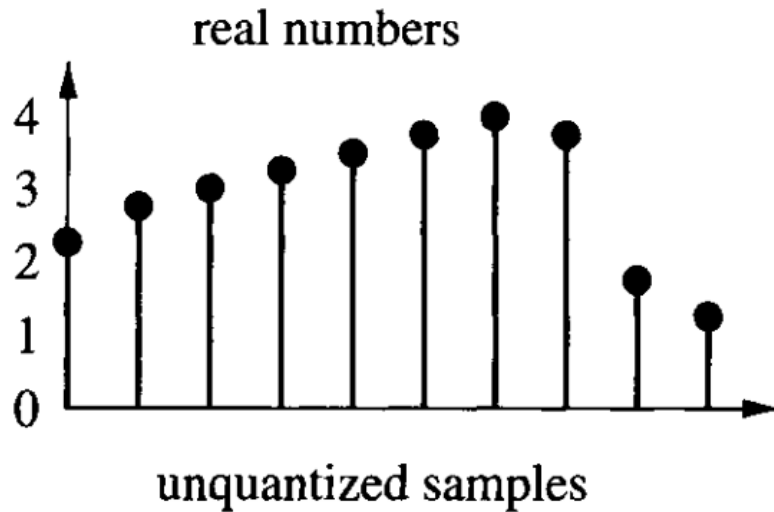
- BTC was compared with the DCT and hybrid coding techniques in the context of high resolution aerial reconnaissance imagery. This study showed that at data rates from 1-3 bits/pixel (monochrome images) BTC performed very favorably compared to the other techniques.
- The use of absolute moment: the mean square performance is better than the standard BTC approach.

5.2 Block Truncation Coding (BTC)

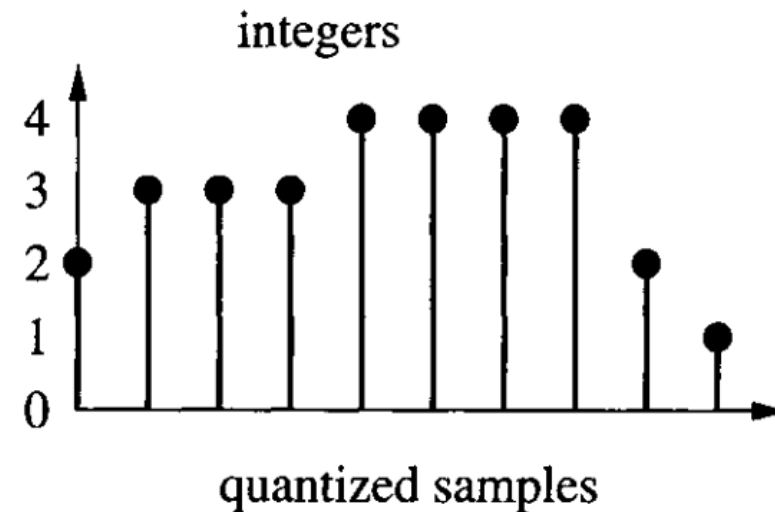
- Implementation
- Discussion and Research
 - Moment Preserving Quantization
 - Absolute moment

5.3 Fundamentals of Vector Quantization

- Quantization is a mapping of a large set of values to a smaller set of values



Infinity precision

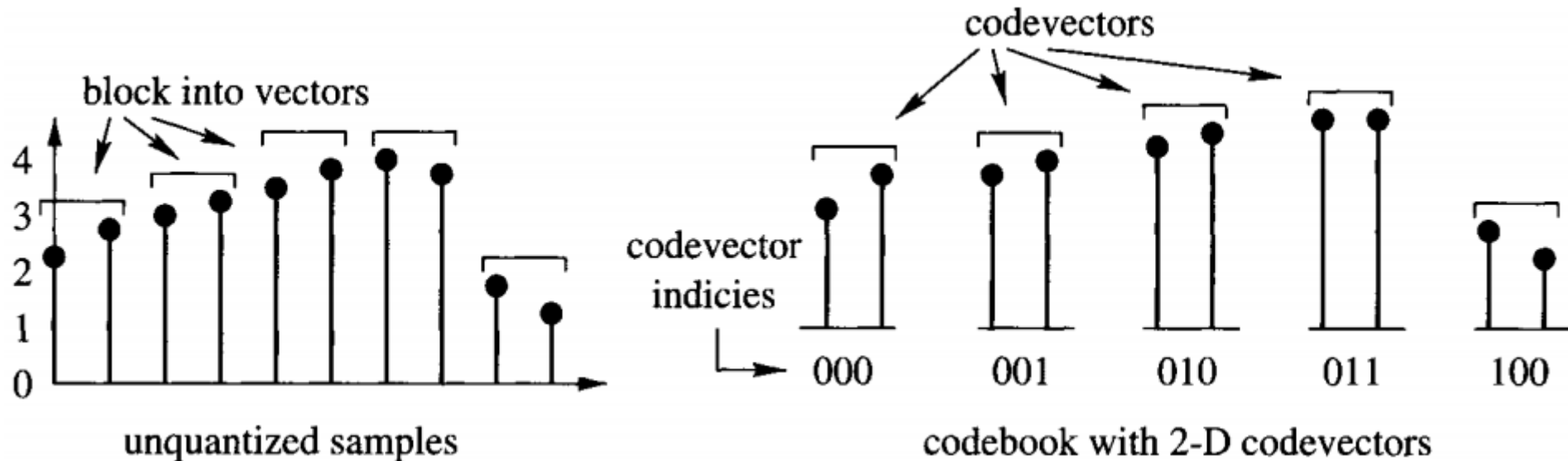


Quantized to integer values

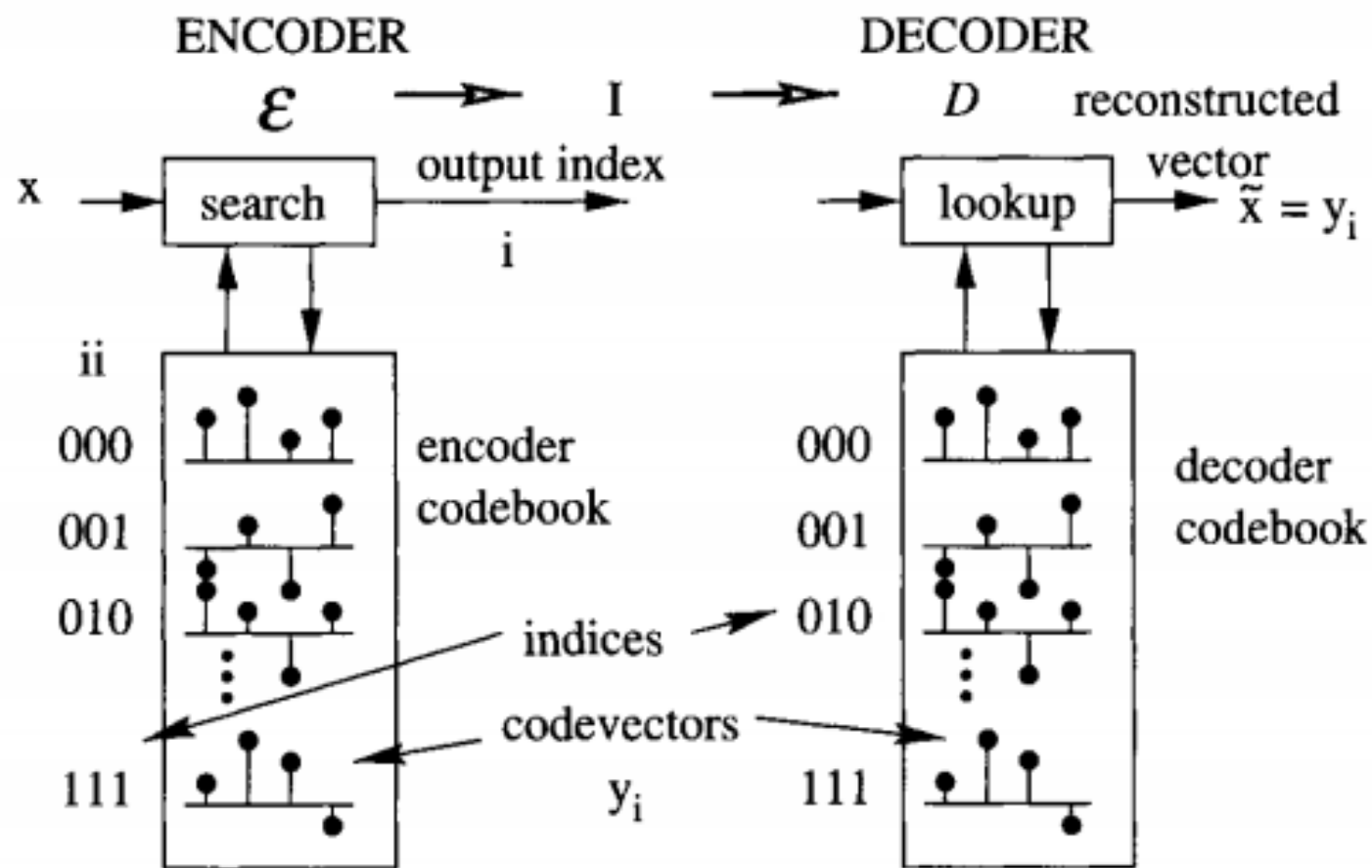
If done wisely, the error introduced by the process can be held to a minimum!

5.3 Fundamentals of Vector Quantization

- Vector quantization (VQ): a mapping from a large set to a smaller set but involves quantizing blocks of samples together



5.3 Fundamentals of Vector Quantization – Encoder and Decoder

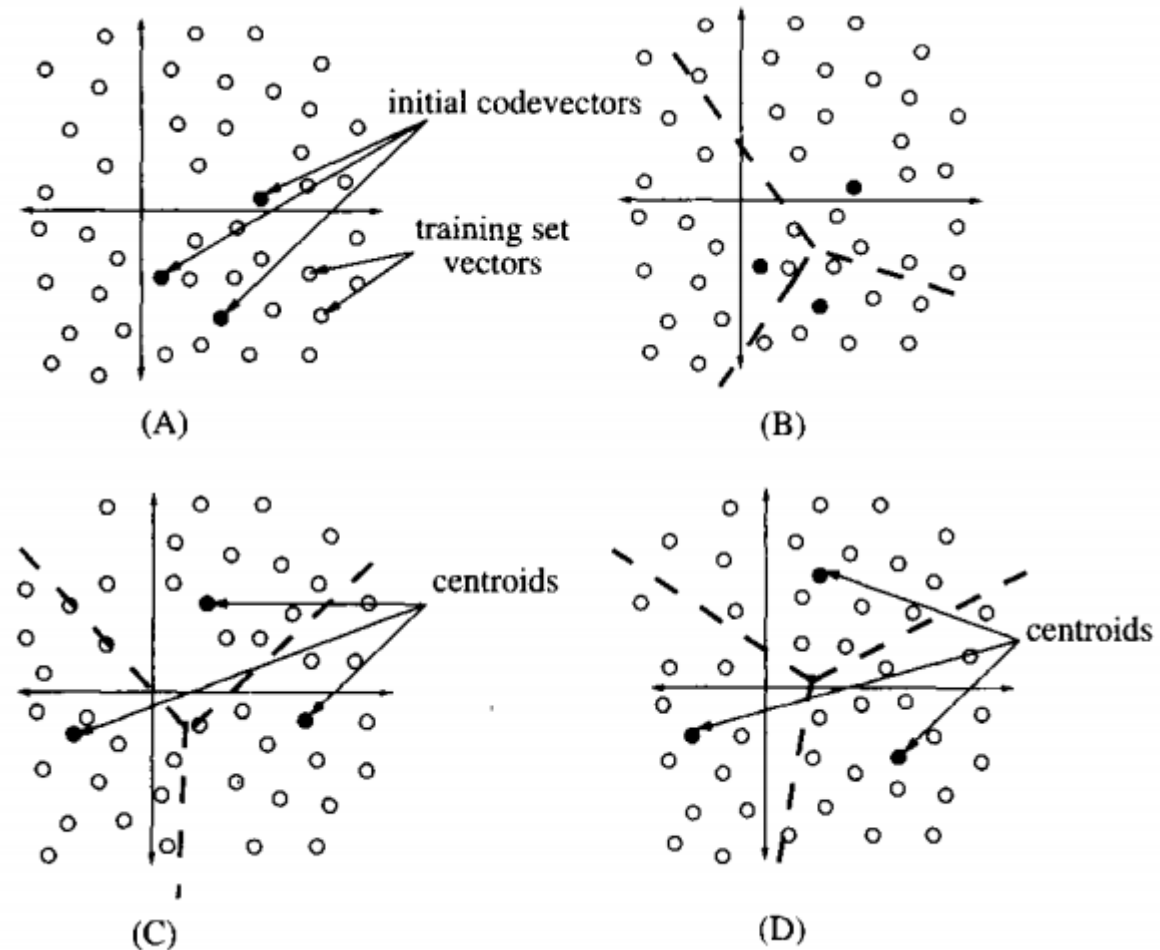


5.3 Fundamentals of Vector Quantization

- The key element in designing a VQ is determining the codebook for a given input source.
- Input source is a large set of representative vectors, called training set
- The most widely cited is the classic iterative method attributed to Linde, Buzo, and Gray (LBG) algorithm
 - Fixed decoder codebook, an optimal encoder partition is the one that satisfies the nearest neighbor rule (map each input vector to the cell producing the smallest distortion)
 - Centroid condition: for a given encoder partition cell, the optimal decoder codeword is the centroid of that cell, where the centroid of cell V_i is the vector $\mathbf{y}^*$ that minimizes $\mathbb{E} \{d(\mathbf{x}, \mathbf{y} | \mathbf{x} \in V_i)\}$ the average distortion in that cell

5.3 Fundamentals of Vector Quantization – Linde, Buzo, and Gray (LBG) algorithm

- First, selects N initial codevectors (selected randomly from the training set)
- Encode all the training vectors using the initial codebook: assigns a subset of the training vectors to each cell defined by the initial code vectors (B)
- The centroid is computed for each cell (C)
- The centroid is used to form an updated codebook until convergence (D)



5.4 Huffman coding

- Huffman coding yields the smallest possible number of code symbols per source symbol
- In terms of information theory, Huffman coding is optimal for a fixed value of n , subject to the constraint that the source symbols be coded one at a time
- The source symbols may be either the intensities of an image or the output of an intensity mapping operation (pixel differences, run lengths, and so on).

5.4 Huffman coding

- The first step in Huffman's approach is to create a series of source reductions by ordering the probabilities of the symbols under consideration, then combining the lowest probability symbols into a single symbol that replaces them in the next source reduction
- The second step in Huffman's procedure is to code each reduced source, starting with the smallest source and working back to the original source.

5.4 Huffman coding

- Illustration of the first and the second steps of Huffman coding

Original source		Source reduction			
Symbol	Probability	1	2	3	4
a_2	0.4	0.4	0.4	0.4	0.6
a_6	0.3	0.3	0.3	0.3	
a_1	0.1	0.1	0.2	0.3	0.4
a_4	0.1	0.1			
a_3	0.06	0.1	0.1	0.1	0.1
a_5	0.04				

Original source			Source reduction			
Symbol	Probability	Code	1	2	3	4
a_2	0.4	1	0.4	1	0.4	1
a_6	0.3	00	0.3	00	0.3	00
a_1	0.1	011	0.1	011	0.2	010
a_4	0.1	0100	0.1	0100	0.1	011
a_3	0.06	01010	0.1	0101	0.1	0101
a_5	0.04	01011				

5.4 Huffman coding

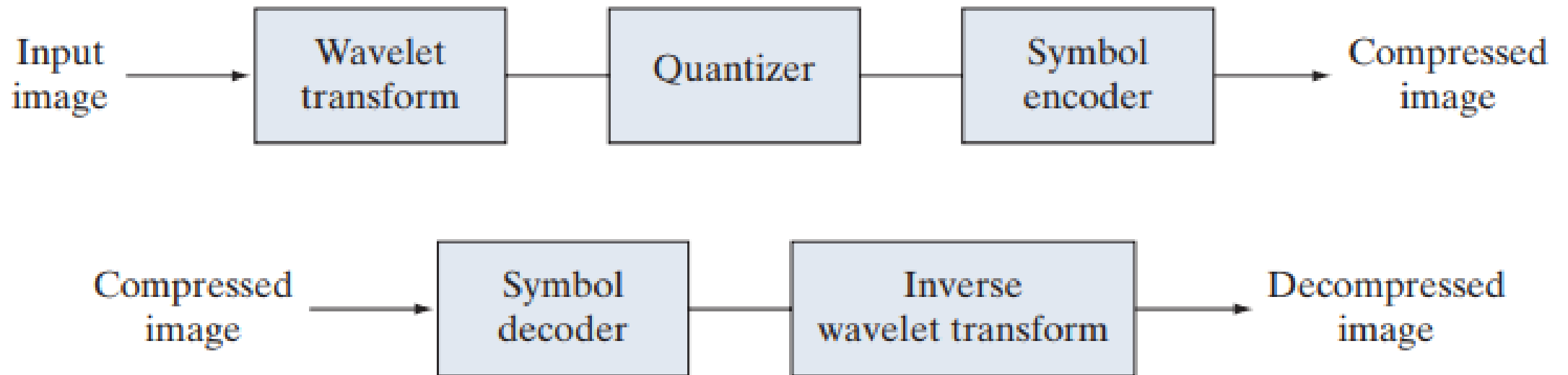
- Discussion: adaptive Huffman coding?

5.5 Wavelet coding

- Wavelet coding is based on the idea that the coefficients of a transform that decorrelates the pixels of an image can be coded more efficiently than the original pixels themselves
- Principle: If the basis functions of the wavelets pack most of the important visual information into a small number of coefficients, the remaining coefficients can be quantized coarsely or truncated to zero with little image distortion.

5.5 Wavelet coding

- Encoder and Decoder



Why there is no subdivision of the original image?

5.5 Wavelet coding –

Some factors affect the compression

- Wavelet function: impact the computational complexity, system's ability to compress and reconstruct images of acceptable error
- Decomposition level selection:
 - Affects computational complexity and reconstruction error
 - The number of operations in the computation of the forward and inverse transforms increases with the number of decomposition levels
 - Quantizing the increasingly lower-scale coefficients that result with more decomposition levels affects increasingly larger areas of the reconstructed image
- Quantizer design
 - Coefficient quantization is the most important factor
 - Widely used: uniform

5.5 Wavelet coding – Some factors affect the compression

- Effectiveness of the quantization can be improved:
 - introducing a larger quantization interval around zero (dead zone)
 - Adapting the size of the quantization interval from scale to scale
- The selected quantization intervals must be transmitted to the decoder with the encoded image bit stream

5.5 Wavelet coding

- Several wavelet functions and compression ability

Wavelet	Filter Taps (Scaling + Wavelet)	Zeroed Coefficients
Haar	2 + 2	33.3%
Daubechies	8 + 8	40.9%
Symlet	8 + 8	41.2%
Biorthogonal	17 + 11	42.1%

Most used

5.5 Wavelet coding

- Example of compressing 512×512 image

Decomposition Level (Scales or Filter Bank Iterations)	Approximation Coefficient Image	Truncated Coefficients (%)	Reconstruction Error (rms)
1	256×256	74.7%	3.27
2	128×128	91.7%	4.23
3	64×64	95.1%	4.54
4	32×32	95.6%	4.61
5	16×16	95.5%	4.63

5.5 Wavelet coding

- Discussion: Adaptive Wavelet transform?

5.6 Arithmetic coding

- Arithmetic coding generates nonblock codes
- An entire sequence of source symbols (or message) is assigned a single arithmetic code word
- The code word itself defines an interval of real numbers between 0 and 1
- As the number of symbols in the message increases, the interval used to represent it becomes smaller
- The number of information units (say, bits) required to represent the interval becomes larger
- Each symbol of the message reduces the size of the interval in accordance with its probability of occurrence

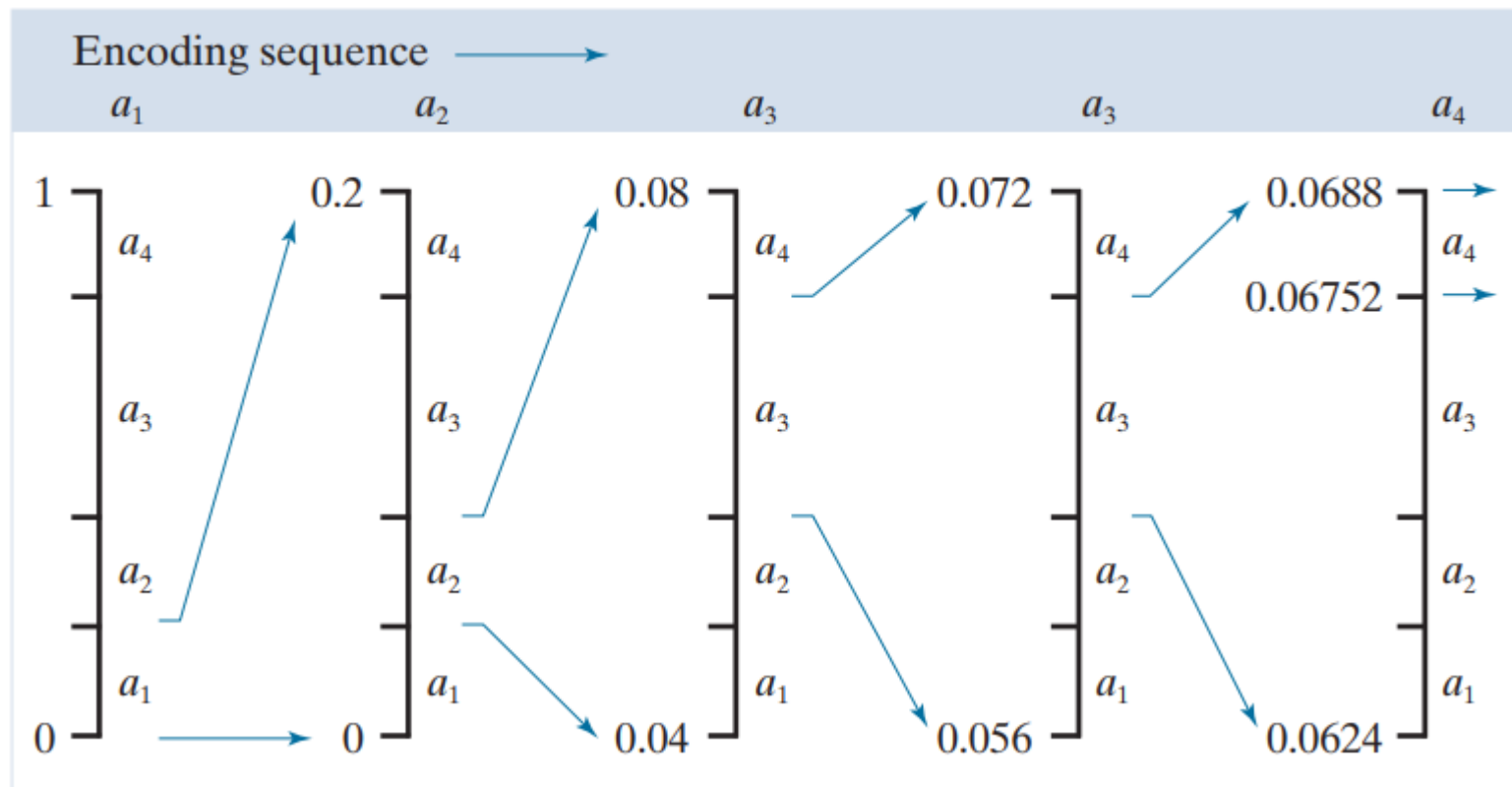
5.6 Arithmetic coding

- An example: encoding a five-symbol sequence or message, $a_1 a_2 a_3 a_3 a_4$ from a four-symbol source is coded

Source Symbol	Probability	Initial Subinterval
a_1	0.2	$[0.0, 0.2)$
a_2	0.2	$[0.2, 0.4)$
a_3	0.4	$[0.4, 0.8)$
a_4	0.2	$[0.8, 1.0)$

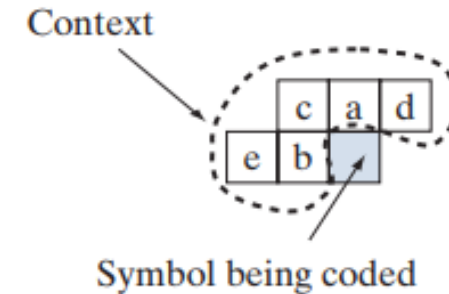
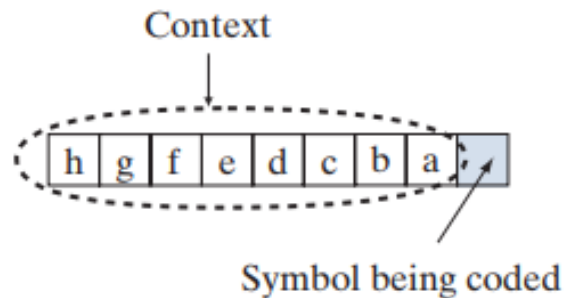
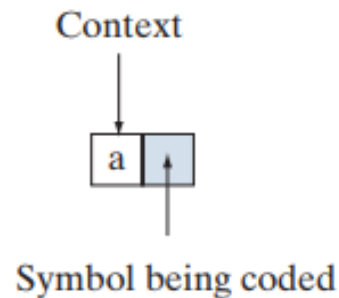
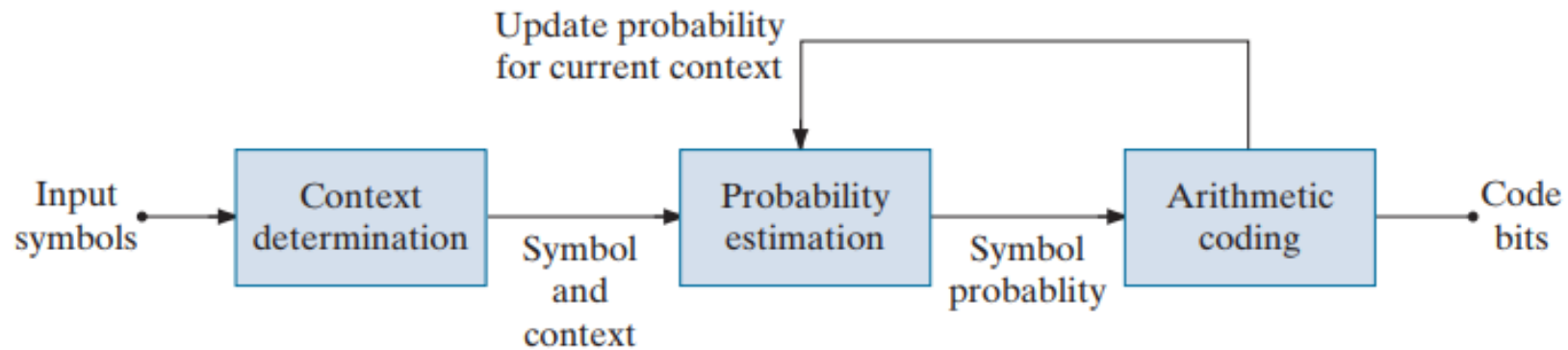
5.6 Arithmetic coding

- Encoding



5.6 Arithmetic coding

- Adaptive context dependent probability estimates



5.6 Arithmetic coding

- Discussion: Which Arithmetic coding or Huffman coding is better in terms of information theory (the number of bits per symbol is smallest)?

5.7 Lossy Image Compression: JPEG and JPEG2000 Standards

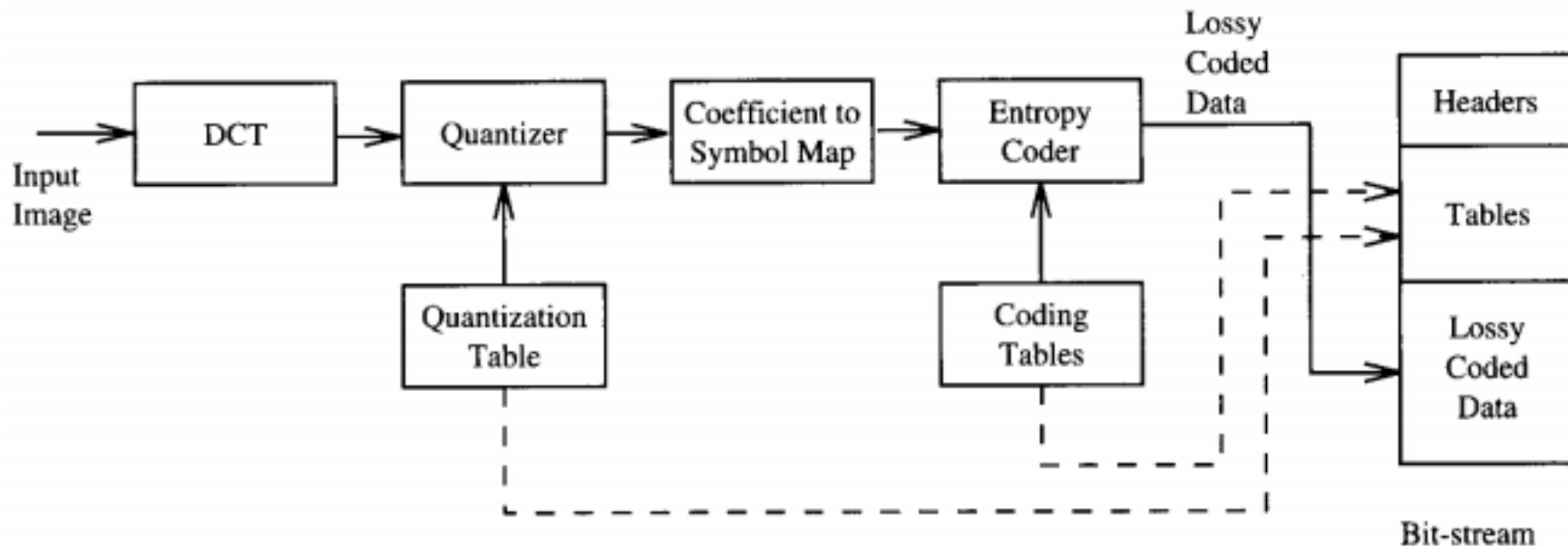
- JPEG (Joint Photographic Experts Group) Standard defines both lossy and lossless compression algorithms
- JPEG lossy compression entails an irreversible mapping of the image to a compressed bit stream, but the standard provides mechanisms for a controlled loss of information.
- Lossy compression produces a bit stream that is usually much smaller in size than that produced with lossless compression.

5.7 Lossy Image Compression: JPEG and JPEG2000 Standards

- The key features of the lossy JPEG standard are
 - Sequential coding encodes quantized DCT coefficients block-by-block in a single scan, moving from left to right and top to bottom.
 - Progressive encoding encodes partial coefficient information in the first scan, with residuals encoded in successive scans.
 - Low-complexity implementations are feasible in both hardware and software.
 - All image types are supported, regardless of source, content, resolution, or color format.
 - A graceful tradeoff between bit rate and quality is possible, except at very low bit rates.
 - A hierarchical mode with multiple resolution levels is allowed.
 - Bit resolutions of 8 to 12 bits are permitted.
 - The JPEG File Interchange Format (JFIF) is recommended for exchanging JPEG bit streams across different platforms.

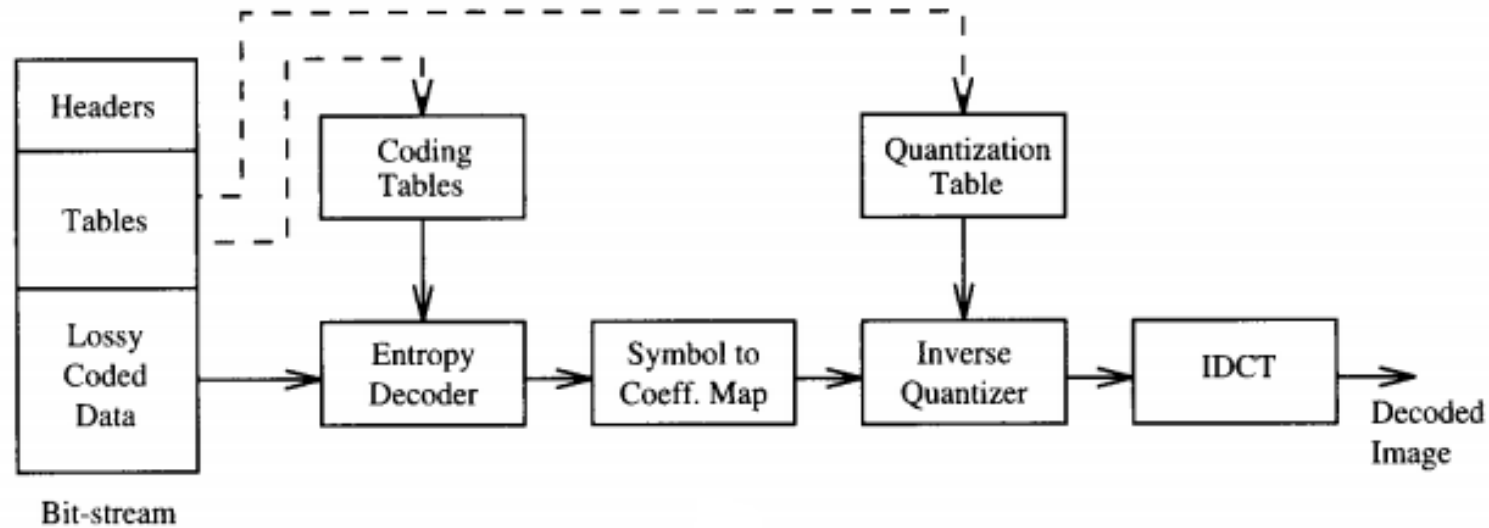
5.7 Lossy Image Compression: JPEG and JPEG2000 Standards

- JPEG encoder



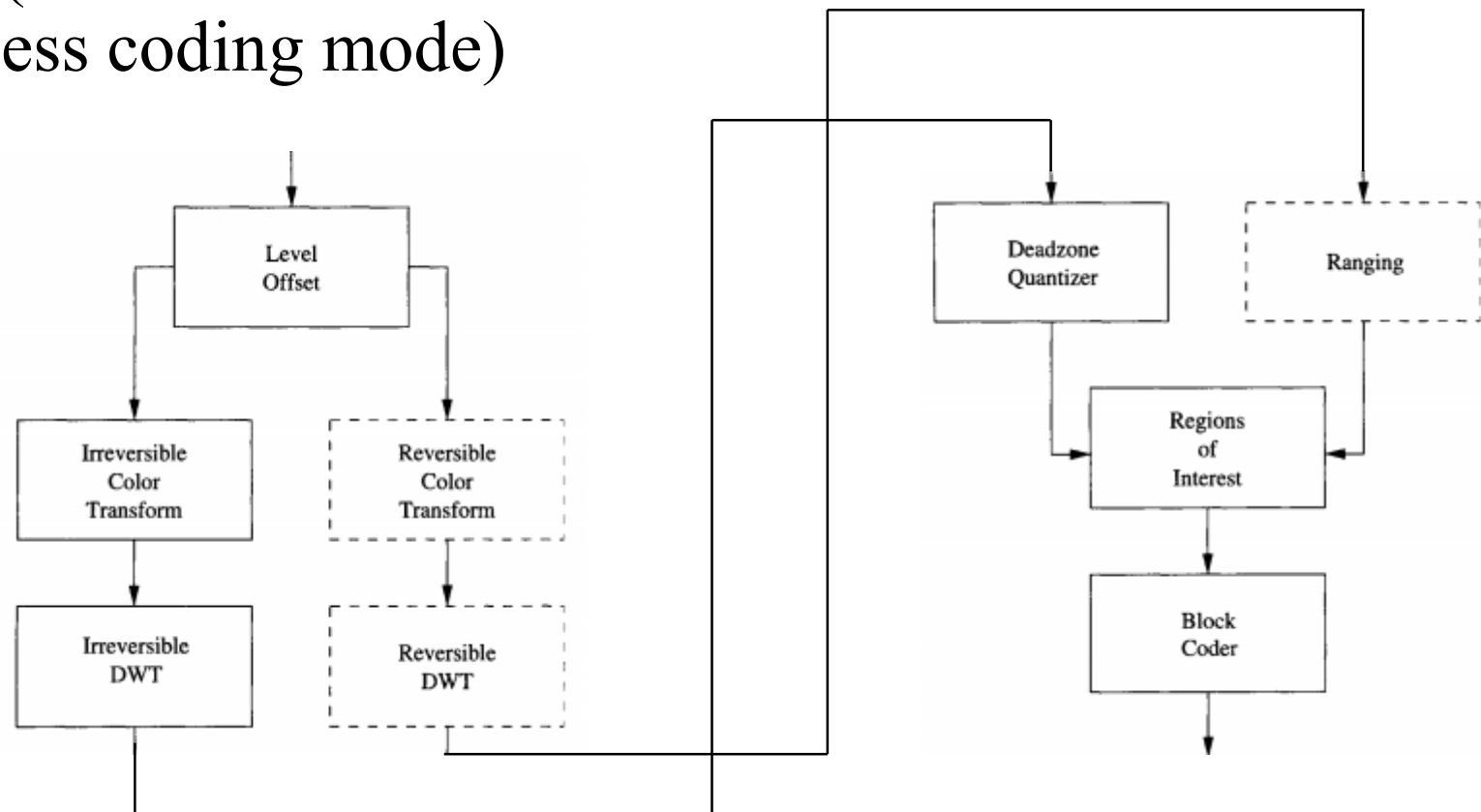
5.7 Lossy Image Compression: JPEG and JPEG2000 Standards

- JPEG decoder



5.7 Lossy Image Compression: JPEG and JPEG2000 Standards

- JPEG2000 encoder (dotted lines corresponds to lossless coding mode)



5.8 Multispectral Image Coding

- Multispectral images, which capture the same spatial region using different imaging modalities (often at various optical wavelengths), require specialized coding algorithms
- Unlike typical two-dimensional images, multispectral images are three-dimensional data sets where the third (spectral) dimension is qualitatively different.
- Standard two-dimensional image compression techniques are usually not suitable for these images.
- Multispectral images are often not intended for human viewing and are typically analyzed by computers, so the criteria for judging decompressed image quality differ from those for two-dimensional images.

5.8 Multispectral Image Coding

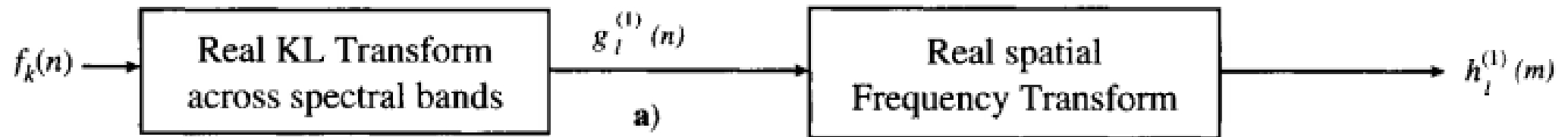
- Application: remotely sensed imagery
- The sensing platform is usually an aircraft or satellite, and the scene being imaged is usually the earth's surface. Since the sensor and the target are so far apart, each pixel in the image can correspond to tens or even hundreds of square meters on the ground.
- Surface features in remotely sensed imagery can be difficult to distinguish with only a few bands of data → a larger number of spectral bands are necessary if a single data set is to be used for multiple purposes.

5.8 Multispectral Image Coding

- Application: medical images
- Magnetic resonance imaging (MRI) can simultaneously measure multiple characteristics of the medium being imaged
- Multispectral medical images can be formed from different medical imaging modalities such as MRI, CT, and x-ray. These multimodal images are useful for identifying and diagnosing medical disorders.

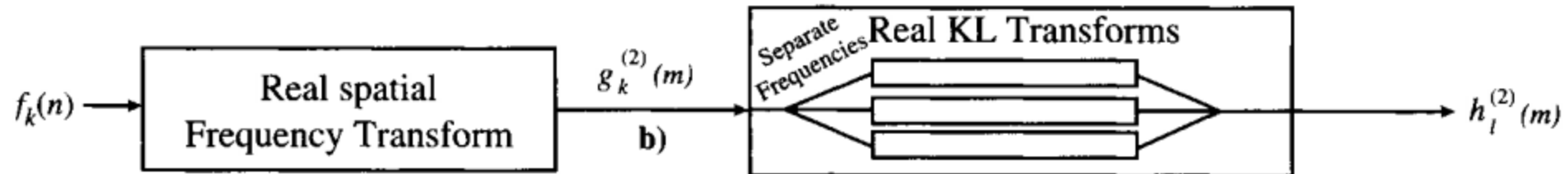
5.8 Multispectral Image Coding

- Transform-based multispectral coders are categorized into three classes
- Spectral-spatial transform: a KL-transform is first applied across the spectral components to decorrelate them. Then each decorrelated component is compressed separately using a transform based image coding method.



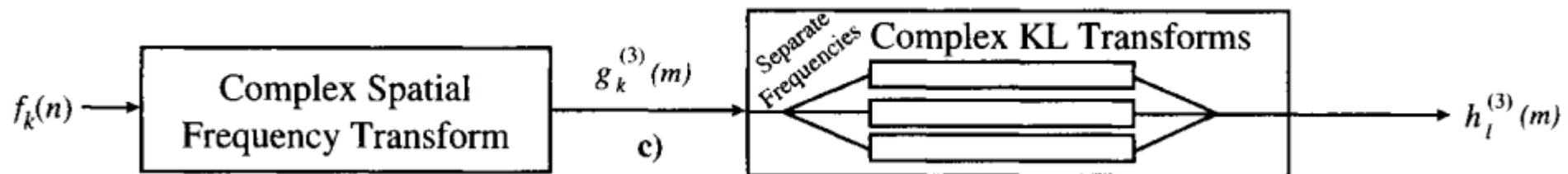
5.8 Multispectral Image Coding

- Transform-based multispectral coders are categorized into three classes
- Spatial-spectral transform: A spatial transform (i.e., block DCT or wavelet transform) is first applied. Then the spectral components of each spatial frequency band are decorrelated using a different KL transform.



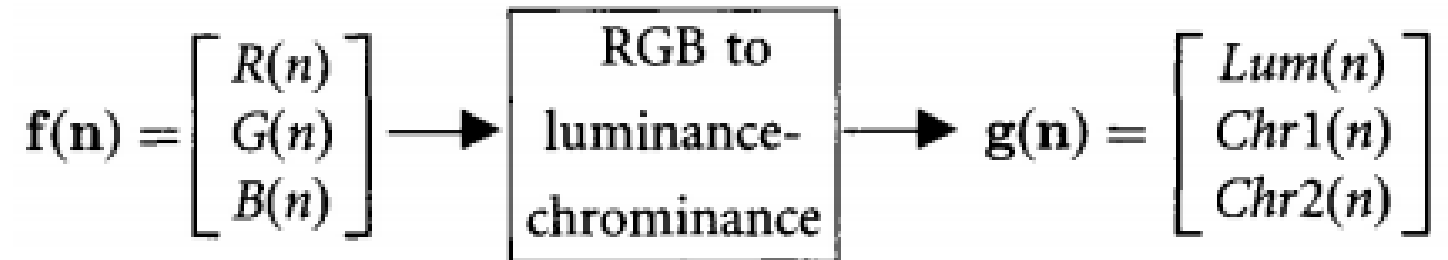
5.8 Multispectral Image Coding

- Transform-based multispectral coders are categorized into three classes
- Complex spatial-spectral transform: If the image planes are not registered, the frequency transforms must be complex to retain phase information between planes. A spatial shift in one image plane relative to another (i.e., misregistration) corresponds to a phase shift in the frequency domain. In order to retain this information, frequency components must be stored as complex numbers.



5.8 Multispectral Image Coding – Lossy compression

- RGB color images
- Before compression color transformation is used to decorrelate the spectral bands visually so they can be treated separately.



5.8 Multispectral Image Coding – Lossless compression

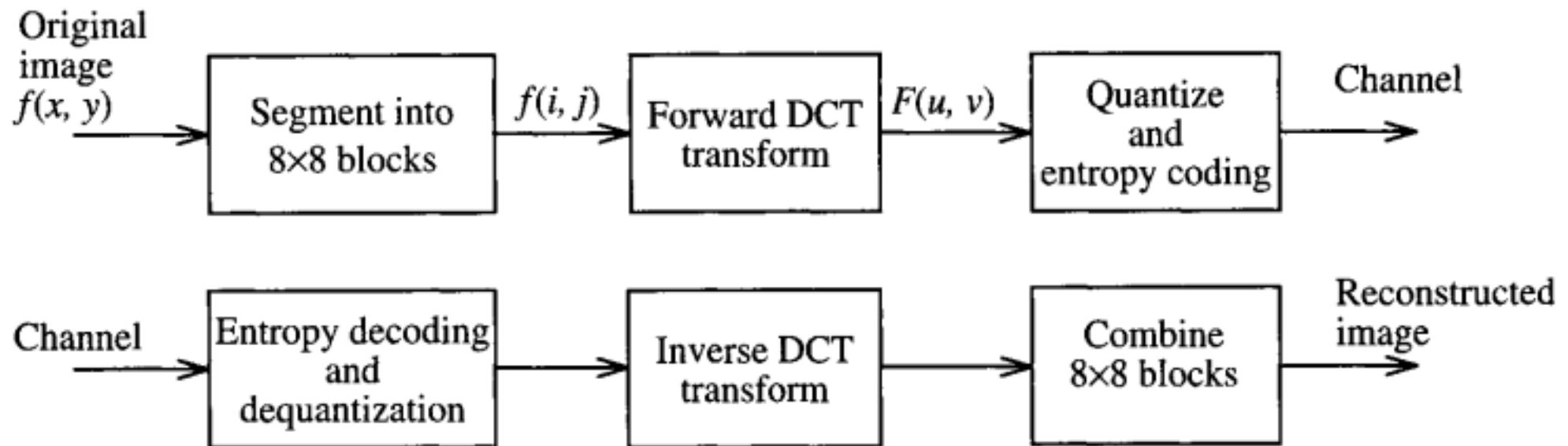
- There are essentially two main approaches used for lossless image compression.
 - The first is the traditional DPCM approach based on prediction followed by context modeling of prediction errors.
 - The second and more recent approach is based on reversible integer wavelet transforms followed by context modeling and coding of transform coefficients.

5.9 Recovery Methods for Postprocessing of Compressed Images

- The objective of postprocessing in a lossy image compression system is to enhance the quality of images produced by the decoder.
- While these systems achieve high compression ratios, they also discard information considered unimportant, leading to distortion in the original image → Postprocessing aims to mitigate this distortion.
- The distortions produced by lossy compression algorithms can be categorized into two classes.
 - "ringing" artifacts: appears at high compression ratios in all transform-based codecs due to the low-pass nature of such systems
 - "blocking" artifact: originates from the independent quantization of the block discrete cosine transform (DCT) coefficients

5.9 Recovery Methods for Postprocessing of Compressed Images

A. The JPEG Discrete Cosine Transform Algorithm



5.9 Recovery Methods for Postprocessing of Compressed Images

A. The JPEG Discrete Cosine Transform Algorithm

1. The image to be encoded is partitioned into contiguous 8×8 -pixel square blocks
2. Each block is transformed by the DCT given by

$$F(u, v) = \frac{C(u)C(v)}{4} \sum_{j=0}^7 \sum_{k=0}^7 f(j, k) \cos \left[\frac{(2j+1)u\pi}{8} \right] \cos \left[\frac{(2k+1)v\pi}{8} \right]$$

where

$$C(u) = \begin{cases} 0 & \text{for } u = 0 \\ 1 & \text{otherwise} \end{cases}$$

$$C(v) = \begin{cases} 0 & \text{for } v = 0 \\ 1 & \text{otherwise} \end{cases}$$

5.9 Recovery Methods for Postprocessing of Compressed Images

3. The DCT coefficients are normalized using a user-defined normalization array $Q(u, v)$, which is the same for all blocks.
4. The normalized and quantized DCT value is given by

$$\tilde{F}(u, v) = \text{round} \left[\frac{F(u, v)}{Q(u, v)} \right]$$

5. The set of quantized DCT coefficients are further encoded using a lossless encoding algorithm such as Huffman encoding
6. At the receiver, after Huffman decoding, the received coefficients are "dequantized" by the operation

$$\hat{F}(u, v) = \tilde{F}(u, v)Q(u, v)$$

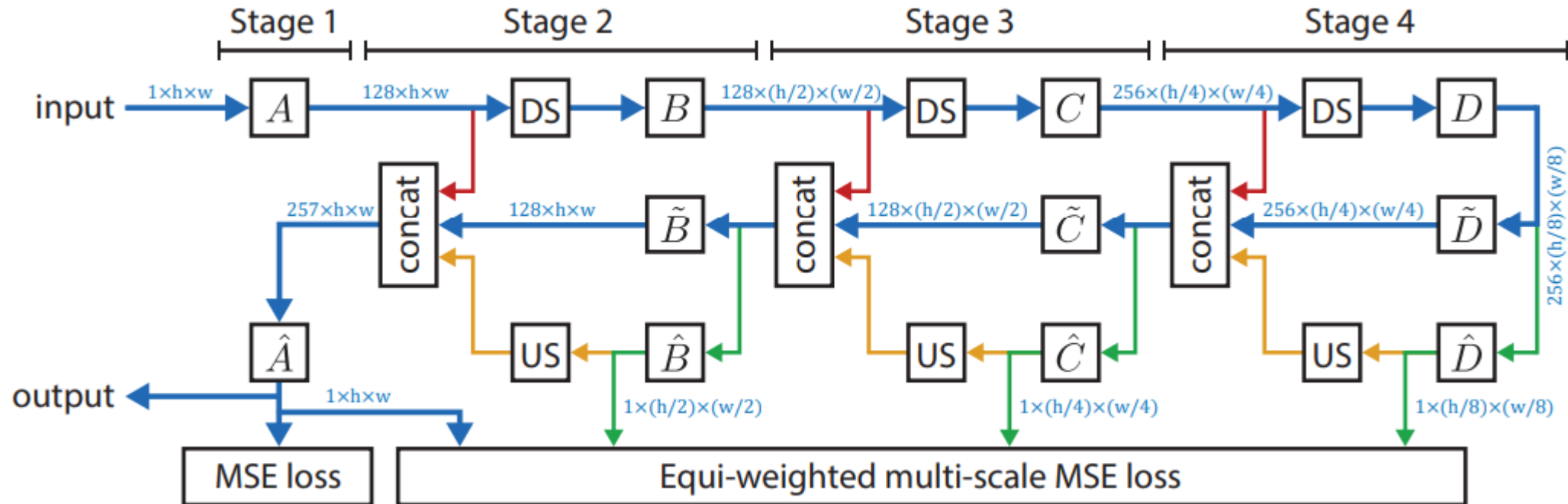
5.9 Recovery Methods for Postprocessing of Compressed Images

7. The dequantized coefficients for each block are used to construct a subimage that is one block of the reconstructed image. The subimage is obtained via the inverse block DCT

$$\hat{f}(j, k) = \frac{1}{4} \sum_{j=0}^7 \sum_{k=0}^7 C(u)C(v)\hat{F}(u, v) \cos \left[\frac{(2j+1)u\pi}{16} \right] \cos \left[\frac{(2k+1)v\pi}{16} \right]$$

5.10 Image compression based on machine learning/deep learning

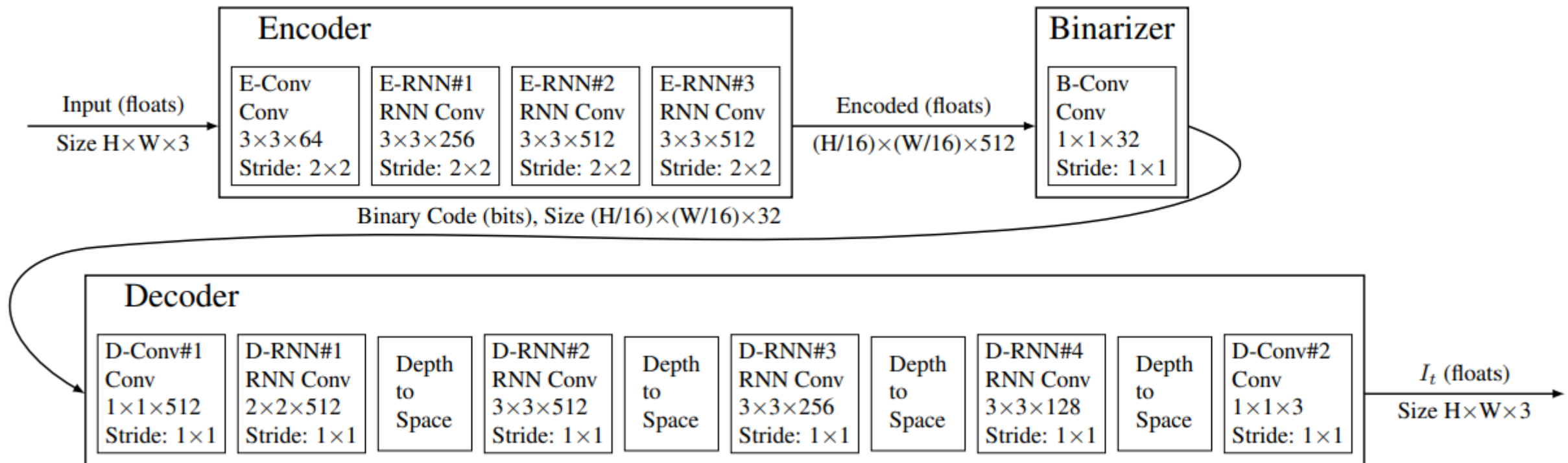
- CNN-based image compression



Cavigelli, L., Hager, P., & Benini, L. (2017). CAS-CNN: A deep convolutional neural network for image compression artifact suppression. 2017 International Joint Conference on Neural Networks (IJCNN). doi:10.1109/ijcnn.2017.7965927

5.10 Image compression based on machine learning/deep learning

- RNN-based image compression



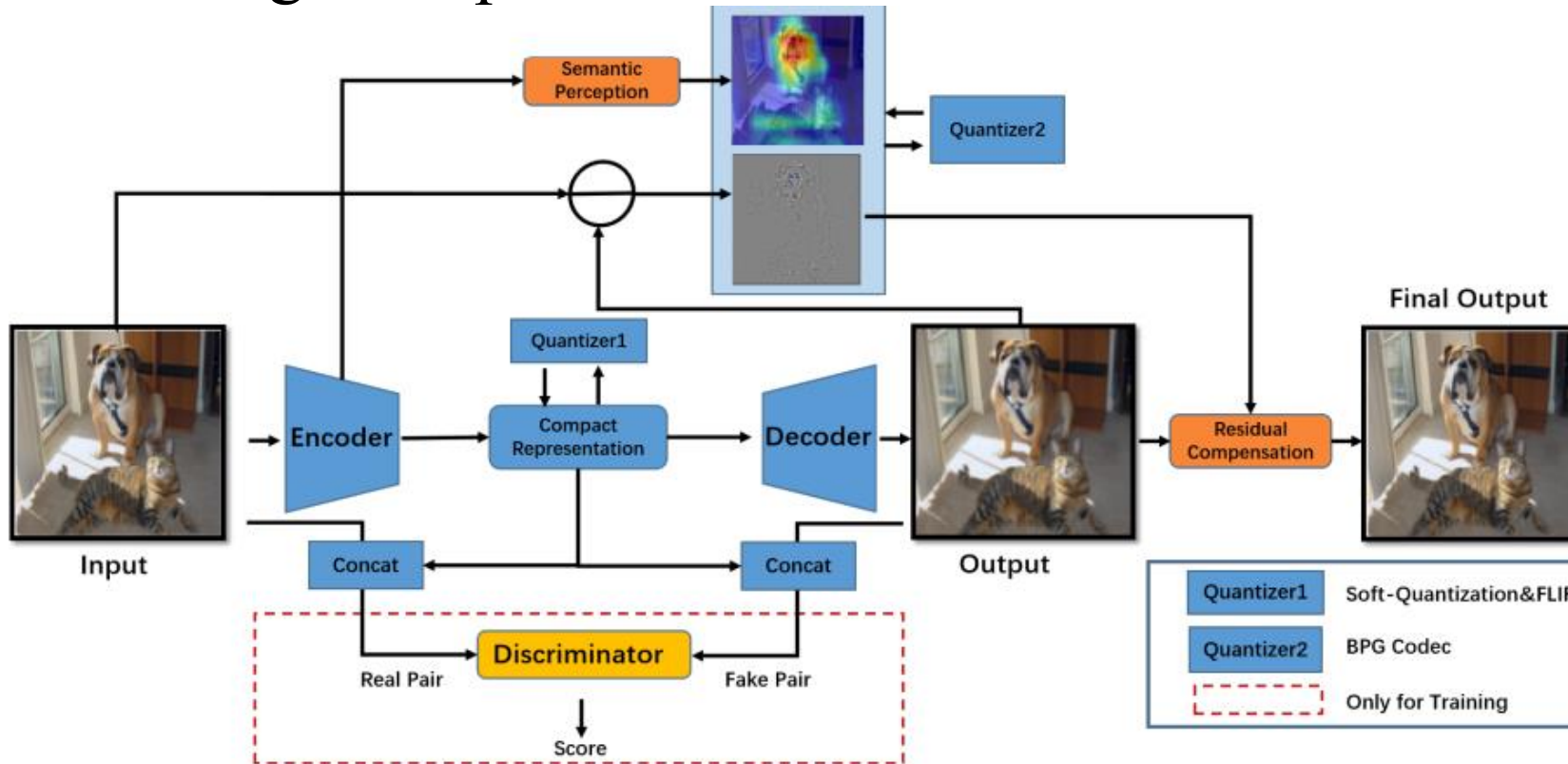
Toderici, G., Vincent, D., Johnston, N., Hwang, S. J., Minnen, D., Shor, J., & Covell, M. (2017).

Full Resolution Image Compression with Recurrent Neural Networks.

2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). doi:10.1109/cvpr.2017.577

5.10 Image compression based on machine learning/deep learning

- GAN-based image compression



Wang, R., Sun, Z., & Kamata, S. (2021). Adaptive Image Compression Using GAN based Semantic-perceptual Residual Compensation. 2020 25th International Conference on Pattern Recognition (ICPR). doi:10.1109/icpr48806.2021.9412655

Summary

- Lossless coding
- Block Truncation Coding (BTC)
- Fundamentals of Vector Quantization
- Huffman coding
- Wavelet Image Compression
- Arithmetic coding
- Lossy Image Compression: JPEG and JPEG2000 Standards
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